Surface developments, true shapes & intersections

GRAPHICS WITH MR KELLY

National 5 · Drawing & spatial awareness · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ identify correct surface developments of prisms, cylinders, pyramids and cones
- ☐ interpret common geometric forms with single cuts

I DO — Worked example

Read how a full-mark answer is built

A cylindrical sweet tube is 40 mm in diameter.

1 Calculate the length of the rectangle in its surface development, to the nearest mm. (1 mark)

Points that earn marks from the marking instructions: one ✓ per point

✓ π × 40 = 126 mm (125.7)

WE DO — Guided practice

Use the starters and word bank to help you

A company prints thousands of card packaging nets for a phone.

2 Describe two ways the production of the packaging could be made more environmentally friendly. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

recycled · possible · sustainable · reduce · environmentally · vegetable · thickness · friendly

Tip: 2 marks = 2 separate, developed points.

YOU DO — On your own

No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) The length of a cylinder's development is…

☐ The diameter ☐ The radius ☐ π × diameter ☐ 2 × height

b) Fitting as many nets as possible onto a sheet…

☐ Increases waste ☐ Reduces waste and cost ☐ Makes folding harder ☐ Is not allowed

c) A cone develops into…

☐ A rectangle ☐ A sector of a circle ☐ A triangle ☐ A square

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4 Key terms: write the letter of the matching description next to each term.

1. **Surface development**	____	**A** A sector with radius equal to the slant height.
2. **Cylinder development**	____	**B** Rectangle of length π × d, plus two circles.
3. **Cone development**	____	**C** The flat net that folds to make a 3D form.

5 Explain what is meant by ‘Glue tab’. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?